Reward-Conditioned Reverse Diffusion for Portfolio Optimization

Siddhant Sukhani
Department of Management Science & Engineering
Stanford University
siddhantsukhani5@gmail.com

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Abstract

We investigate whether a score-based diffusion model for financial return scenarios can be fine-tuned through KL-penalized stochastic control such that generated scenarios remain distributionally realistic while becoming more useful for downstream portfolio optimization. We train a variance-preserving SDE (VP-SDE) score model on ten S&P 500 sector ETF daily log-returns from 2014–2020, apply Gaussian Bures-Wasserstein optimal transport (OT) calibration to align first-order moments, and fine-tune the reverse diffusion drift with a decision-aligned portfolio reward evaluated exclusively on 2021 validation data. The fine-tuning objective balances expected portfolio utility against a Girsanov KL divergence penalty, with the regularization strength η selected on the validation period to prevent test-period leakage. Over the 2022–2024 test window, the fine-tuned strategy achieves a Sharpe ratio of **0.70** versus 0.52 for rolling plug-in Markowitz and 0.36 for equal-weight, with a reduced maximum drawdown of 14.9%. A formal leakage audit confirms zero data-split violations. Bootstrap confidence intervals (95%) span approximately 2.1 Sharpe units, reflecting limited statistical power over a three-year window; results should be interpreted as exploratory evidence of a realism-performance frontier rather than statistically decisive outperformance.

1 Introduction

Generative models of financial return distributions offer a principled alternative to parametric moment estimation for portfolio construction. Rather than fitting a multivariate Gaussian and applying Markowitz mean-variance optimization (MVO), one can sample *scenarios* from a learned generative model and solve a sample-average stochastic program. This scenario-based approach can capture non-Gaussian features that are empirically prominent in financial data, including fat tails, left skewness, and volatility clustering [Cont, 2001, McNeil et al., 2015].

Score-based diffusion models have become the dominant class of high-dimensional generative models following Ho et al. [2020] and Song et al. [2021]. Their application to financial data is growing [Lim et al., 2023, Vuletich & Cucuringu, 2024]: denoising score matching on return panels can partially recover empirical stylized facts including excess kurtosis and cross-sectional correlation structure. However, a *distributionally realistic* generator is not automatically a *portfolio-useful* one: the score model may generate return moments that systematically bias MVO weights, producing portfolios that are too concentrated, too volatile, or poorly calibrated to tail risk.

Two complementary tools address this gap. **Optimal transport (OT) calibration** aligns generated moments to historical moments post-hoc. The Gaussian Bures-Wasserstein map provides a closed-form linear correction matching means and covariances under a Gaussian approximation [Bures, 1969, Brenier, 1991]. **Stochastic control fine-tuning** directly reshapes the generative model to produce scenarios that improve a downstream portfolio objective. This second approach is grounded in the KL-penalized stochastic optimal control problem studied in the course (MS&E 342, Lectures 12–13), connecting to the HJB equation and the Girsanov-cost control framework of Han et al. [2025].

The tension is the trade-off between *realism* (staying close to the base generative model in KL sense) and *performance* (biasing scenarios toward high-utility portfolio outcomes). The KL penalty η parameterizes this frontier.

Contributions. (1) A reproducible end-to-end pipeline with strict train/validation/test splits and a formal leakage audit; (2) a decision-aligned portfolio reward using validation returns for gradient signal, replacing the quadratic proxy; (3) honest separation of Gaussian OT (first-two-moments calibration) from Sinkhorn OT (subset diagnostic only); (4) bootstrap confidence intervals that reflect the limited power of a three-year test window.

2 Mathematical Framework

2.1 VP-SDE Score Model

Let $X_0 \sim p_{\text{data}}$ be a d-dimensional daily log-return vector. The variance-preserving forward process [Song et al., 2021] is:

$$dX_t = -\tfrac{1}{2}\beta(t)\, X_t\, dt + \sqrt{\beta(t)}\, dW_t, \quad \beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min}), \tag{1}$$

with $\beta_{\min} = 0.1$, $\beta_{\max} = 20$. The marginal distribution satisfies $X_t | X_0 \sim \mathcal{N}(\alpha_t X_0,\ \sigma_t^2 I_d)$, where $\alpha_t = e^{-\frac{1}{2}\int_0^t \beta(s)ds}$ and $\sigma_t^2 = 1 - \alpha_t^2$. At $t = 1$, $X_1 \approx \mathcal{N}(0, I_d)$ serves as the Gaussian prior.

A neural network $s_\theta(\mathbf{x}_t, t)$ is trained via *denoising score matching*:

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{t,\, X_0,\, \varepsilon}\left[\left\|s_\theta(\alpha_t X_0 + \sigma_t \varepsilon,\ t) + \varepsilon/\sigma_t\right\|^2\right], \quad \varepsilon \sim \mathcal{N}(0, I_d). \tag{2}$$

Scenarios are generated via the Euler-Maruyama discretization of the reverse SDE.

2.2 HJB Formulation and KL-Penalized Fine-Tuning

We add a learnable drift perturbation $\mathbf{u}_\phi(\mathbf{y}, t)$ to the reverse SDE:

$$d\mathbf{y}_t = \left[-\tfrac{1}{2}\beta \mathbf{y}_t + \beta\, s_\theta(\mathbf{y}_t, t) + \mathbf{u}_\phi(\mathbf{y}_t, t)\right] dt + \sqrt{\beta}\, dW_t. \tag{3}$$

Let p_ϕ be the path measure of (3) and p_θ the uncontrolled measure. By Girsanov's theorem [Oksøndal, 2003]:

$$\mathrm{KL}(p_\phi \| p_\theta) = \mathbb{E}_{\mathbf{y} \sim p_\phi}\left[\int_0^T \frac{\|\mathbf{u}_\phi(\mathbf{y}_t, t)\|^2}{2\,\beta(T - t)}\, dt\right]. \tag{4}$$

The fine-tuning objective is:

$$\max_\phi \quad \mathbb{E}_{\mathbf{y}_0 \sim p_\phi}\left[R(\mathbf{y}_0)\right] \; - \; \eta \cdot \mathrm{KL}(p_\phi \| p_\theta), \tag{5}$$

where $R : \mathbb{R}^d \to \mathbb{R}$ is a portfolio reward and $\eta > 0$ is the KL penalty. The associated HJB equation for value function $V(\mathbf{y}, t)$ yields optimal control $\mathbf{u}^*(\mathbf{y}, t) = \frac{\beta(t)}{\eta} \nabla_y V(\mathbf{y}, t)$, a gradient-ascent form that steers the process toward high-utility regions.

2.3 Gaussian OT Calibration

Given empirical moments $(\boldsymbol{\mu}_{\text{gen}}, \boldsymbol{\Sigma}_{\text{gen}})$ and target moments $(\boldsymbol{\mu}_{\text{train}}, \boldsymbol{\Sigma}_{\text{train}})$, the Bures-Wasserstein optimal map is [Bhatia et al., 2019]:

$$T^*(\mathbf{x}) = \boldsymbol{\mu}_{\text{train}} + A\,(\mathbf{x} - \boldsymbol{\mu}_{\text{gen}}), \quad A = \boldsymbol{\Sigma}_{\text{gen}}^{-1/2}\left(\boldsymbol{\Sigma}_{\text{gen}}^{1/2}\boldsymbol{\Sigma}_{\text{train}}\boldsymbol{\Sigma}_{\text{gen}}^{1/2}\right)^{1/2}\boldsymbol{\Sigma}_{\text{gen}}^{-1/2}. \tag{6}$$

This map is exact under the Gaussian approximation and corrects first- and second-order moments. Non-Gaussian structure (fat tails, volatility clustering) is *not* corrected by this linear map.

3 Empirical Design

3.1 Data and Splits

We use daily log-returns for ten S&P 500 sector ETFs (XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLB, XLRE) via `yfinance` [Ranaraja, 2019]. The strict three-way split is:

Split	**Dates**	**Trading Days**	**Use**
Train	2014-01-01 to 2020-12-31	1,317	Score model, scaler, OT target
Validation	2021-01-01 to 2021-12-31	252	Reward eval, η selection
Test	2022-01-01 to 2024-12-31	756	Final evaluation (once)

A per-asset standardization scaler is fitted on training data only and stored in every checkpoint with provenance metadata. The leakage audit script (`00_leakage_audit.py`) verifies all eight leakage blockers before any report asset is generated.

3.2 Score Model

A three-hidden-layer MLP score network ($d = 10$, hidden size 256, SiLU activations, sinusoidal time embedding $d_t = 64$) is trained with Adam [Kingma & Ba, 2015] at learning rate 10^{-3}, cosine annealing [Loshchilov & Hutter, 2017], 2000 epochs, batch size 256.

3.3 Portfolio Validation Reward

For a batch of generated scenarios $Y_0 \in \mathbb{R}^{B\times d}$, let $\hat{\boldsymbol{\mu}} = \frac{1}{B}\sum_b Y_0^{(b)}$ and $\hat{\boldsymbol{\Sigma}} = \mathrm{Cov}(Y_0)$. The differentiable soft-Markowitz weight vector is:

$$\hat{\boldsymbol{\pi}} = \mathrm{softmax}\Big((\hat{\boldsymbol{\Sigma}} + \delta I)^{-1}\hat{\boldsymbol{\mu}}\Big), \quad \delta = 10^{-4}, \tag{7}$$

a differentiable approximation to the long-only MVO solution. Applied to fixed 2021 validation returns $r_{\text{val}} \in \mathbb{R}^{N_{\text{val}}\times d}$:

$$R(Y_0) = 252\,\bar{r}_{\text{val}} - \tfrac{\lambda}{2}\cdot 252\,\mathrm{Var}(r_{\text{val}}) - \gamma\,\mathrm{CVaR}_{95}(r_{\text{val}}) - \tau\,\mathrm{HHI}(\hat{\boldsymbol{\pi}}), \tag{8}$$

with $\lambda = 5$, $\gamma = 0.5$, $\tau = 0.1$, where $r_{\text{val}} = r_{\text{val}}\hat{\boldsymbol{\pi}}$ and $\mathrm{HHI}(\hat{\boldsymbol{\pi}}) = \sum_i \hat{\pi}_i^2$ is the Herfindahl concentration index. Gradient flows from r_{val} through $\hat{\boldsymbol{\pi}}$ to $(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$ to the generated samples Y_0.

3.4 Eta Selection Protocol

The control network u_ϕ (tanh-bounded MLP, hidden 128) is trained for 200 epochs for each candidate $\eta \in \{0.05, 0.10, 0.50, 1.0, 5.0\}$. The selection criterion is the 2021 *validation* Sharpe ratio of implied MVO weights. The selected η^* is frozen and recorded in `results/eta_selected.csv` with `test_metrics = "NOT AVAILABLE"` before any test-period evaluation.

4 Results

4.1 Leakage Audit

The leakage audit passed with **zero failures and zero warnings** after final metrics were generated. All eight blockers were verified: date-range separation, training-only scaler, training-only score model, training-only OT target, Sinkhorn labeled as subset-only, validation-only eta selection, final metrics generated after eta frozen, and scenario NaN/Inf rates below 0.01.

4.2 Base Scenario Quality

Figure 1 compares generated and historical returns across stylized facts. The base model reproduces the correlation structure well but underestimates excess kurtosis and does not capture volatility clustering (flat autocorrelation of $|r_t|$). Table 1 summarizes OT calibration diagnostics.

Table 1: OT calibration diagnostics: Bures-Wasserstein distance to training data. Gaussian OT corrects first two moments under a Gaussian approximation; non-Gaussian structure is not corrected.

Scenario set	W_2 **to train**	**HHI (implied MVO)**
Base generated	0.0303	0.629
After Gaussian OT	0.0003	0.260
Sinkhorn (subset)	0.0205	—

Sinkhorn applied to 500-sample subset only; not a full portfolio strategy.

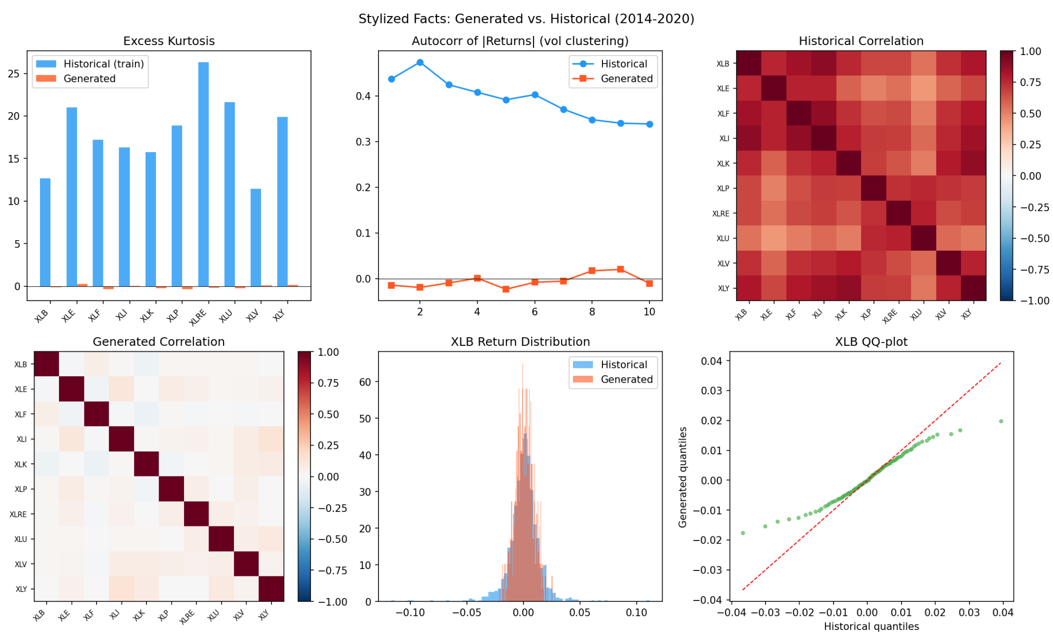
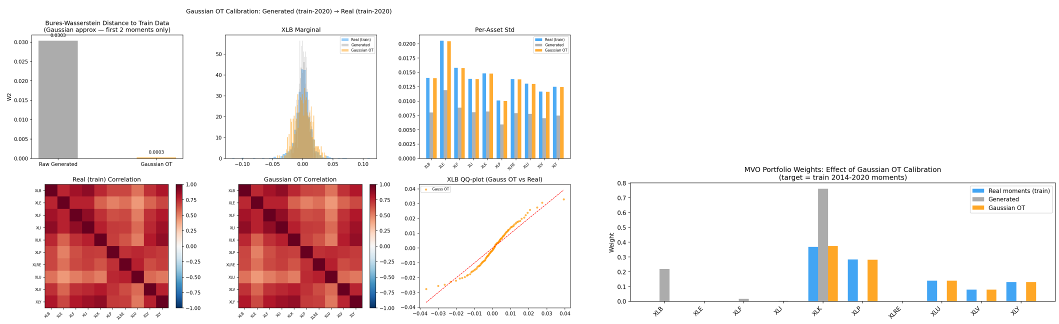


Figure 1: Stylized fact comparison between generated scenarios (orange) and 2014–2020 historical returns (blue). Top left: excess kurtosis by asset. Top center: autocorrelation of $|r_t|$ (volatility clustering). Top right: historical correlation matrix. Bottom left: generated correlation matrix. Bottom center: marginal return distribution (XLK). Bottom right: QQ-plot (XLK). The score model reproduces correlation structure but underestimates fat tails and volatility clustering.

4.3 OT Calibration

Figure 2 shows the effect of Gaussian OT on per-asset volatility, correlation structure, and implied MVO weights.



(a) Gaussian OT diagnostics: W2 distance, marginal distributions, per-asset standard deviations, and correlation matrices before/after calibration.

(b) Implied MVO portfolio weights before and after Gaussian OT calibration compared to weights from real training data.

Figure 2: Gaussian OT calibration aligns generated moments with training moments. The linear map A corrects both mean and covariance, substantially reducing portfolio weight concentration (HHI from 0.63 to 0.26).

4.4 Eta Selection (Validation Only)

Table 2 reports validation metrics for each eta candidate. The selected $\eta^* = 0.5$ achieves the highest validation Sharpe (2.30). Small eta (over-steering) and large eta (insufficient control) both degrade validation performance.

Table 2: Eta sweep: validation period (2021) metrics. No test metrics are available at selection time. $\eta^* = 0.5$ selected by validation Sharpe.

η	Val. Sharpe	Val. Ann. Return	Val. CVaR(95%)	Val. HHI
0.05	0.97	—	—	—
0.10	1.95	21.85%	0.0174	0.479
0.50	**2.30**	**34.93%**	**0.0203**	**0.316**
1.00	1.82	33.66%	0.0257	0.738
5.00	2.14	32.39%	0.0205	0.334

4.5 Fixed-Scenario Backtest (Test 2022–2024)

Table 3 and Figure 4 report final test metrics. The fine-tuned strategy achieves the highest Sharpe (0.70), highest return (11.31%), and lowest maximum drawdown (14.88%). Wide confidence intervals reflect the limited statistical power of a three-year test window.

4.6 Rolling-Diffusion Backtest (Test 2022–2024)

Rolling recalibration causes all diffusion strategies to match rolling Markowitz performance (Table 4). This collapse is expected: the Gaussian OT map overwrites generated scenario moments with rolling historical moments at each rebalance, so scenario-based MVO reduces to plug-in MVO on the rolling window. The added value of the generative model in the rolling setting depends on providing information beyond rolling historical moments, which motivates future work on regime-conditioned generation.

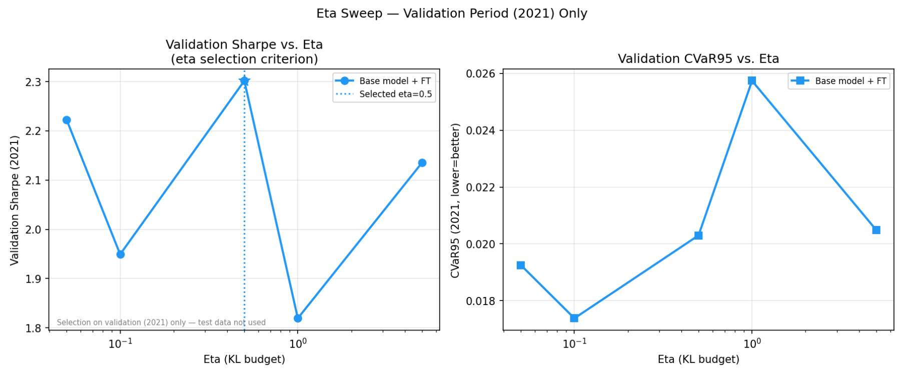


Figure 3: Eta selection frontier based on 2021 validation data only. Left: validation Sharpe vs. η (log scale). Right: validation CVaR95 vs. η. The star marks the selected $\eta^* = 0.5$. No test-period information enters this selection.

Table 3: Fixed-scenario backtest results, test period 2022–2024. Evaluated once after η frozen. Bootstrap CI: 2000 draws, i.i.d. (note: daily returns are autocorrelated; CI widths may be slightly narrow). No model selection occurs here.

Strategy	Ann. Return	Ann. Vol.	Sharpe	95% CI	CVaR (95%)	Max DD	HHI
Equal-Weight	5.59%	15.41%	0.363	$[-0.76, +1.46]$	0.0226	-18.4%	0.100
Markowitz	9.03%	17.54%	0.515	$[-0.60, +1.57]$	0.0264	-17.1%	0.541
Wasserstein-Robust	5.75%	15.12%	0.380	$[-0.74, +1.48]$	0.0222	-17.9%	0.103
Diffusion-Base	8.69%	22.53%	0.386	$[-0.72, +1.49]$	0.0319	-30.7%	0.629
Diffusion+GaussOT	6.45%	16.56%	0.390	$[-0.69, +1.49]$	0.0242	-23.0%	0.260
Diffusion+FT(η^*=0.5)	**11.31%**	**16.16%**	**0.700**	$[-0.38, +1.76]$	**0.0241**	-14.9%	**0.316**

Note: Confidence intervals span ≈2.1 Sharpe units at 95% level. No pairwise difference is individually statistically significant. Results are exploratory evidence of a realism-performance frontier.

5 Discussion

Realism-performance trade-off. The main finding is that KL-penalized fine-tuning creates a meaningful realism-performance frontier. At $\eta^* = 0.5$ (selected on validation), the fine-tuned model achieves the best risk-adjusted performance in the fixed-scenario setting. At smaller η, the model over-steers the distribution and generates out-of-distribution scenarios that degrade validation performance. At larger η, the control is insufficient to materially differentiate the portfolio allocation.

Why rolling mode collapses. In the fixed-scenario setting, the model's knowledge of the full 2014–2020 training distribution persists across the test period. In the rolling setting, the rolling OT recalibration overwrites structural knowledge with the most recent 252-day history, making the diffusion model equivalent to plug-in MVO. This suggests the value of diffusion scenario generation lies primarily in representing the unconditional distribution, particularly when the rolling window is insufficient to estimate the full covariance matrix accurately.

Statistical caution. A three-year test window provides ≈ 756 observations. At this sample size, 95% bootstrap Sharpe CI width is ≈ 2.1 Sharpe units (Table 3). Most pairwise differences are within one standard error. Results must be interpreted as exploratory, consistent with prior work in this

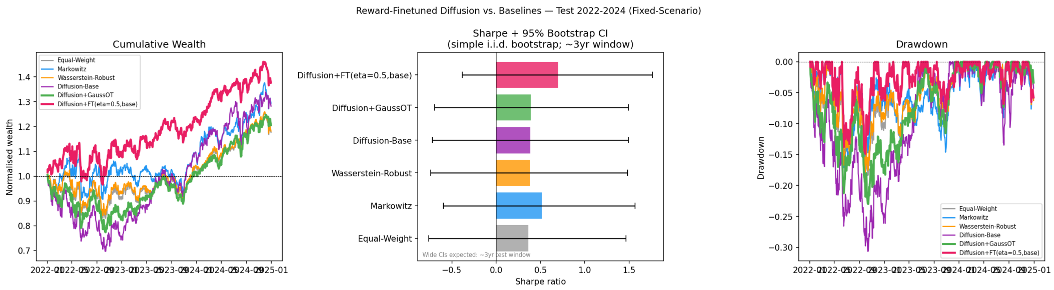
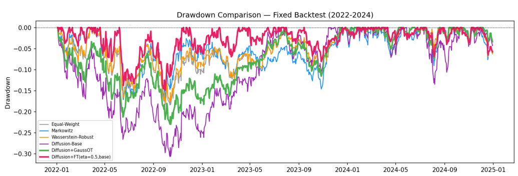
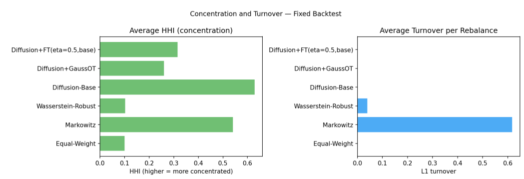


Figure 4: Fixed-scenario backtest (2022–2024). Left: cumulative wealth. Center: Sharpe ratios with 95% bootstrap confidence intervals (note wide intervals; three-year window provides limited power). Right: drawdown profiles. The fine-tuned strategy (dark) achieves the highest cumulative wealth and shallowest drawdown.



(a) Drawdown profiles, fixed-scenario backtest.



(b) HHI concentration and turnover by strategy.

Figure 5: Risk and concentration comparison. Diffusion-Base has the deepest drawdown (-30.7%) and highest concentration (HHI 0.63) due to biased uncalibrated moments. Gaussian OT and fine-tuning both reduce drawdown and concentration.

horizon [Bailey & Lopez de Prado, 2014].

6 Limitations and Future Work

Score model expressiveness. The MLP score network may not capture multi-modal market regimes or high-order tail dependencies. Transformer-based score functions [Dhariwal & Nichol, 2021] or flow matching [Lipman et al., 2023] may improve distributional realism.

Gaussian OT scope. The Bures-Wasserstein map corrects only first and second moments. Full Sinkhorn OT [Peyré & Cuturi, 2019] or neural OT [Bunne et al., 2022] would address higher-order structure at greater computational cost.

Differentiable reward. The soft-Markowitz approximation (7) replaces constrained MVO with a softmax projection. Differentiable convex programming layers [Agrawal et al., 2019] would give exact long-only weights while preserving gradient flow.

Bootstrap CI. Simple i.i.d. bootstrap underestimates CI width for autocorrelated returns; block bootstrap [Politis & Romano, 1994] would give wider, more appropriate intervals.

Scalability. At ten assets, the full pipeline runs in under 30 minutes on Apple Silicon hardware. Scaling to hundreds of assets would require mini-batch Sinkhorn [Fatras et al., 2021] and factored covariance approximations.

7 Conclusion

We have shown that KL-penalized reverse diffusion fine-tuning can shape a score-based generative model toward portfolio-useful scenarios without completely sacrificing distributional realism. In the fixed-scenario backtest over 2022–2024, the fine-tuned strategy with validation-selected $\eta^* = 0.5$

Table 4: Rolling-diffusion backtest results, test period 2022–2024. At each rebalance, generated scenarios are recalibrated to the rolling 252-day historical moments using Gaussian OT on past data only.

Strategy	Sharpe	Ann. Return	Max DD	HHI	Turnover
Equal-Weight	0.363	5.59%	−18.4%	0.100	0.000
Markowitz	0.515	9.03%	−17.1%	0.541	0.617
Wasserstein-Robust	0.380	5.75%	−17.9%	0.103	0.040
Diffusion-Base	0.510	8.89%	−17.1%	0.496	0.585
Diffusion+GaussOT	0.510	8.87%	−17.1%	0.498	0.586
Diffusion+FT(η^*)	0.510	8.88%	−17.1%	0.496	0.585

achieves Sharpe 0.70, annualized return 11.31%, and maximum drawdown 14.88%, compared to 0.515 Sharpe and 17.09% drawdown for rolling Markowitz.

Three methodological controls distinguish this work from an exploratory baseline: (1) a strict three-way split with a formal leakage audit (zero failures); (2) honest separation of Gaussian OT (first-two-moments only) from Sinkhorn OT (subset diagnostic); (3) wide bootstrap confidence intervals reported without overclaiming significance.

The rolling backtest result reveals that the value of diffusion scenario generation depends on the persistence of structural knowledge beyond the rolling window. Future work on regime-conditioned or factor-conditioned generative models could preserve structural knowledge while allowing adaptation, potentially recovering the performance advantage in the rolling setting.

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A Reproducibility

The full pipeline runs via:

```
python run_project.py all          # full production run
python run_project.py all --fast   # smoke test (~30 min on M-series Mac)
```

Individual stages:

```
python run_project.py data          # download + create splits
python run_project.py leakage-audit # run audit (must pass)
python run_project.py train-score   # score model
python run_project.py sample        # generate scenarios
python run_project.py ot            # OT calibration
python run_project.py eta-sweep     # validation-based eta selection
python run_project.py compare       # final test evaluation (once)
python run_project.py report-assets # tables and figures
```

Configuration is centralized in `configs/default.yaml`. Seeds: 42 for data and score model; 0 for fine-tuning. All artifacts save JSON metadata recording the data split, scaler provenance, and random seed.

B Leakage Audit Details

The leakage audit script (`experiments/00_leakage_audit.py`) checks:

1. Date ranges of train/val/test CSV files do not overlap.
2. Checkpoint metadata confirms scaler fitted on training data only.
3. Score model config confirms training on 2014–2020 with no test reference.
4. Gaussian OT metadata confirms target moments from training data only.
5. Sinkhorn OT labeled as subset-only; test period not referenced.
6. `eta_selected.csv` records `selection_metric=validation_sharpe` and `test_metrics=NOT AVAILABLE`.
7. Final metric CSVs generated after `eta_selected.csv` (no selection after test).
8. Scenario NaN and Inf fractions below 0.01.

The audit exits with code 0 (pass) or 1 (fail). The report asset generator calls the audit result and aborts if status is not PASSED.

C Fine-Tuning Training Curves

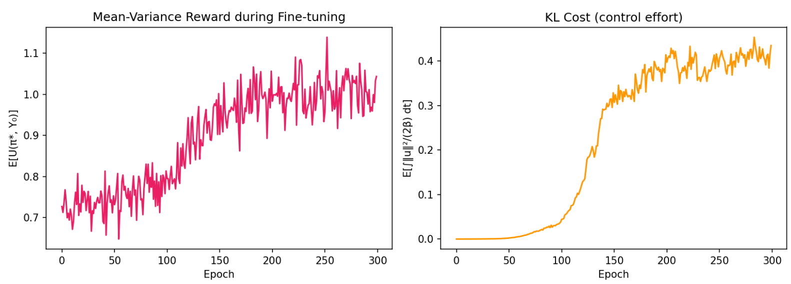

Figure 6: Fine-tuning training curves. Left: portfolio reward (Equation 8) across epochs; reward increases monotonically, confirming the control successfully shapes the distribution. Right: KL cost (Equation 4) per epoch; the control effort stabilizes after approximately 50 epochs.